

Pentolite

Pentolite is a composite high explosive used for military and civilian purposes, e.g., warheads and booster charges. It is made of pentaerythritol tetranitrate (PETN) phlegmatized with trinitrotoluene (TNT) by melt casting.^[1]

The most common military variety of pentolite (designated "Pentolite 50/50") is a mixture of 50% PETN and 50% TNT.^[1] (Unlike other compound explosives, the number before the slash is the mass percentage of TNT and the second number is the mass percentage of PETN.) This 50:50 mixture has a density of 1.65 g/cm³ and a detonation velocity of 7400 m/s.

Pentolite is a common explosive for cast boosters for the blasting work (as in mining).^{[2][3]} Civilian pentolite may contain a lower percentage of PETN, e. g. around 2% ("Pentolite 98/2"), 5% ("Pentolite 95/5") or 10% ("Pentolite 90/10"). These civilian pentolites have a detonation velocity of about 7,800 metres per second.

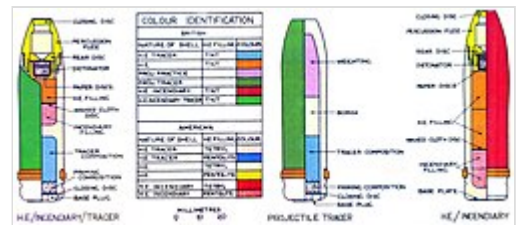
References

1. B. M. Dobratz & P. C. Crawford, *LLNL Explosives Handbook*, Lawrence Livermore National Laboratory, UCRL-52997, Change 2, January 31, 1985.
2. Kennedy, Bruce A. (1990). *Surface Mining, Second Edition* (<https://books.google.com/books?id=qJJrYnpT2pYC&pg=PA547>). SME. p. 547. ISBN 978-0-87335-102-7.
3. Rustan, Agne (1998). *Rock Blasting Terms and Symbols: A Dictionary of Symbols and Terms in Rock Blasting and Related Areas like Drilling, Mining and Rock Mechanics* (<https://books.google.com/books?id=idsQOABCipMC&pg=PA33>). CRC Press. p. 33. ISBN 978-90-5410-441-4.

External links

- Additional information in re Pentolite boosters (<https://web.archive.org/web/20061015165245/http://vetrivelexplosives.com/sunPenta.htm>)

Retrieved from "<https://en.wikipedia.org/w/index.php?title=Pentolite&oldid=1310183545>"



Cross-sectional view of Oerlikon 20 mm cannon shells (dating from circa 1945) showing color code for pentolite filling